

**Compounds of Transitional elements**

181. Steel becomes soft and pliable by  
 (a) Annealing  
 (b) Nitriding  
 (c) Tempering  
 (d) Case hardening
182. Most stable oxidation state of iron is  
 (a) +2 (b) +3  
 (c) -2 (d) -3
183. Nickel steel contain % of Ni  
 (a) 1 – 5% (b) 3 – 5%  
 (c) 6 – 5% (d) 8 – 5%
184. Permanent magnet is made from  
 (a) Cast iron (b) Steel  
 (c) Wrought Iron (d) All of these
185. In nitriding process of steel  
 (a) Steel is heated in an atmosphere of ammonia  
 (b) Steel is made red hot and then cooled  
 (c) Steel is made red hot and then plunged into oil for cooling  
 (d) None of these
186. Iron on reacting with carbon give  
 (a)  $FeC$  (b)  $Fe_2C$   
 (c)  $Fe_3C$  (d)  $FeC_2$
187. Iron loses magnetic property at  
 (a) Melting point  
 (b) 1000K  
 (c) Curie point  
 (d) Boiling point
188. Heat treatment alters the properties of steel due to  
 (a) Chemical reaction on heating  
 (b) Partial rusting  
 (c) Change in the residual energy  
 (d) Change in the lattice structure due to differential rate of cooling
189. Pure conc.  $HNO_3$  makes iron passive as the surface is covered with protective layer of  
 (a)  $Fe_2O_3$  (b)  $FeO$   
 (c)  $Fe_3O_4$  (d)  $Fe(NO_3)_3$
190. Red hot iron absorbs  $SO_2$  giving the product  
 (a)  $FeS + O_2$   
 (b)  $Fe_2O_3 + FeS$   
 (c)  $FeO + FeS$   
 (d)  $FeO + S$
191. If steel is heated to a temperature well below red heat and is then cooled slowly, the process is called  
 (a) Tempering (b) Hardening  
 (c) Softening (d) Annealing
192. In smelting of iron, which of the following reactions takes place in Blast furnace at  $400^\circ C - 600^\circ C$   
 (a)  $CaO + SiO_2 \rightarrow CaSiO_3$   
 (b)  $2FeS + 3O_2 \rightarrow 2Fe + SO_2$   
 (c)  $FeO + SiO_2 \rightarrow FeSiO_3$   
 (d)  $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$



193. Soil containing both *Al* and *Fe* is called  
(a) Laterite (b) Bauxite  
(c) Pedalfers (d) Clay
194. German silver is an alloy of  
(a) Copper, zinc and nickel  
(b) Copper and silver  
(c) Copper, zinc and tin  
(d) Copper, zinc and silver
195. Iron is rendered passive by the action of  
(a) Conc.  $H_2SO_4$   
(b) Conc.  $H_3PO_4$   
(c) Conc.  $HCl$   
(d) Conc.  $HNO_3$
196. Iron sheets are galvanized by depositing a coating of **or**. In galvanisation, iron surface is coated with  
(a) Zinc (b) Tin  
(c) Chromium (d) Nickel
197. Chemical formula of rust is  
(a)  $FeO$  (b)  $Fe_3O_4$   
(c)  $Fe_2O_3 \cdot xH_2O$  (d)  $FeO \cdot xH_2O$
198. Heating steel to bright redness and then cooling suddenly by plunging it into oil or water, makes it  
(a) Hard and pliable  
(b) Soft and pliable  
(c) Soft and brittle  
(d) Hard and brittle
199. Which of the following is found in body  
(a) *Pb* (b) *Fe*  
(c) *Cd* (d) *Al*
200. Which of the following pairs of elements might form an alloy  
(a) Zinc and lead  
(b) Iron and mercury  
(c) Iron and carbon  
(d) Mercury and platinum

